Week 4 Lab Assistant Preparation Guide

Technology Projects - Cisco Packet Tracer Networking Lab

1. What Week 4 is about

Week 4 is about basic computer networking using Cisco Packet Tracer.

There are two main activities:

1. Basic LAN setup - students connect multiple PCs to a switch, manually assign IP addresses, and test communication using ping.
2. DHCP and DNS setup - students create a small network with a server, switch, and PCs, then use DHCP for automatic IP assignment and DNS for name resolution.

The main goal is for students to understand what a LAN is, why devices need IP addresses, what a switch does, how devices test connectivity, why DHCP is useful, and why DNS is useful.

2. Relevant repo folders/files

Week 4 is inside:

```
week04/
```

It has two activities:

```
week04/network1/
week04/network2/
```

Activity 1: Basic LAN

```
week04/network1/README.md
week04/network1/01_summary.md
week04/network1/02_topology.md
week04/network1/03_IP_configuration.md
week04/network1/04_connectivity.md
week04/network1/05_reflection.md
```

This activity asks students to build a LAN with:

```
5 PCs
1 switch
Copper Straight-Through cables
Manual IP addresses
Ping tests
```

Device	IP Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.11	255.255.255.0
PC2	192.168.1.12	255.255.255.0
PC3	192.168.1.13	255.255.255.0
PC4	192.168.1.14	255.255.255.0

Activity 2: DHCP and DNS

```
week04/network2/README.md
week04/network2/01_objective.md
week04/network2/02_topology_setup.md
week04/network2/03_server.md
week04/network2/04_clients.md
week04/network2/05_testing.md
week04/network2/06_reflection.md
```

This activity asks students to build a network with:

```
1 server
1 switch
3 PCs
DHCP enabled on server
DNS enabled on server
PCs set to DHCP
Ping tests using hostnames
```

Setting	Value
Server IP	192.168.2.2
Subnet mask	255.255.255.0
Default gateway	192.168.2.1
DHCP start IP	192.168.2.100
DHCP max users	50
DNS server	192.168.2.2

3. What students are expected to do

Activity 1: Basic LAN

1. Open Cisco Packet Tracer.
2. Add 5 PCs.
3. Add 1 switch, probably a 2960 switch.
4. Connect each PC to the switch using Copper Straight-Through cables.
5. Manually assign IP addresses to each PC.
6. Test connection using ping.
7. Try pinging an unused IP address such as ping 192.168.1.15.
8. Write their answers in the journal files.
9. Add screenshots where requested.

They should understand that all PCs are in the same network:

```
192.168.1.0/24
```

Activity 2: DHCP and DNS

1. Open Cisco Packet Tracer.
2. Add 1 server, 1 switch, and 3 PCs.
3. Connect all devices to the switch.
4. Give the server a static IP address: 192.168.2.2.

5. Enable DHCP on the server.
6. Configure the DHCP pool.
7. Set each PC to use DHCP instead of static IP.
8. Check that PCs receive IP addresses such as 192.168.2.100, 192.168.2.101, and 192.168.2.102.
9. Enable DNS on the server.
10. Add DNS records that match the PC IP addresses.
11. Test DNS using ping.
12. Fill in the journal files and add screenshots.

4. What you should personally understand before the lab

LAN

A LAN means Local Area Network. A simple explanation: A LAN is a small network where nearby devices can communicate with each other, like computers in one classroom, office, or building.

Switch

A switch connects devices inside the same local network. It is like a central meeting point. All computers plug into it, and it forwards messages to the correct computer. Students may think the switch gives IP addresses. It does not.

IP address

An IP address identifies a device on a network. It is like a house address for a computer. If another computer wants to send it data, it needs to know its address.

Subnet mask

The subnet mask tells the computer which part of the IP address is the network part. With 255.255.255.0, devices usually need the first three numbers to match if they want to communicate directly.

Ping

Ping checks whether another device is reachable. It is like asking, Are you there? If the other device replies, the connection works.

DHCP

DHCP automatically gives IP addresses to devices. It saves us from typing IP addresses manually.

DNS

DNS converts names into IP addresses. It is like a phonebook. Instead of remembering 192.168.2.100, we can use a name like library_pc1.

5. Common student mistakes and how to fix them

Using the wrong cable

Fix: Use Copper Straight-Through cable. Connect PC FastEthernet0 to a switch FastEthernet port.

Forgetting to assign IP addresses

Fix: In Activity 1, go to PC -> Desktop -> IP Configuration and enter the correct IP address and subnet mask.

Putting PCs in different networks

Fix: Make sure all Activity 1 PCs use 192.168.1.x with subnet mask 255.255.255.0.

Duplicate IP addresses

Fix: Give each device a unique IP. Two computers cannot have the same address in the same network.

Pinging a non-existing IP and thinking the lab is broken

Fix: Ping to 192.168.1.15 should fail if no device has that address. This is intentional.

DHCP server is not turned on

Fix: Go to Server -> Services -> DHCP and make sure the service is ON, then save the pool.

Forgetting to click Save/Add in DHCP or DNS

Fix: Packet Tracer often requires clicking Save for DHCP and Add for DNS records.

PCs are still using static IP instead of DHCP

Fix: Go to PC -> Desktop -> IP Configuration and select DHCP, then wait a few seconds.

DNS name mismatch

Fix: The name students ping must exactly match the name added in the DNS server.

DNS record points to the wrong IP

Fix: Use ipconfig /all first, then create DNS records based on the actual assigned IPs.

Default gateway confusion

Fix: The DHCP settings include 192.168.2.1, but no router is shown. For this same-network lab, the gateway is not the main focus.

6. Commands/tools/concepts you should be ready to explain

Packet Tracer tools

```
End Devices -> PC
End Devices -> Server
Network Devices -> Switches -> 2960
Connections -> Copper Straight-Through
```

PC IP configuration

```
PC -> Desktop -> IP Configuration

Manual/static:
IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
```

```
DHCP:
Select DHCP
```

Command Prompt

```
PC -> Desktop -> Command Prompt

ping 192.168.1.11
ping 192.168.2.2
ping library_pc1
ipconfig /all
```

Server DHCP

```
Server -> Services -> DHCP

Service: ON
Pool name: serverPool
Default Gateway: 192.168.2.1
DNS Server: 192.168.2.2
Start IP Address: 192.168.2.100
Subnet Mask: 255.255.255.0
Maximum Number of Users: 50
```

Server DNS

```
Server -> Services -> DNS

library_pc1 -> 192.168.2.100
library_pc2 -> 192.168.2.101
library_pc3 -> 192.168.2.102

Test:
ping library_pc1
```

7. Physical or software setup to check before the session

Software setup

- Cisco Packet Tracer is installed on lab machines.
- Students can open Packet Tracer without login problems.
- The repo is accessible to students.
- Students know where to write their journal answers.
- Students know how to take screenshots and where to save them.

Repo setup

The repo expects screenshots such as:

```
images/topology.png
images/ip-config-pc0.png
images/ping-pc0.png
```

If the images folder does not exist, students may need to create it manually inside each activity folder.

```
mkdir images
```

Lab assistant setup

- Prepare one completed Activity 1 Packet Tracer file.
- Prepare one completed Activity 2 Packet Tracer file.
- Prepare screenshots of working IP configuration and ping tests.
- Prepare a simple diagram on paper or whiteboard for both activities.

8. Simple explanations you can give to beginners

What is a network?

A network is when computers are connected so they can send data to each other.

What is a LAN?

A LAN is a small local network, like computers in one room, one office, or one building.

What is a switch?

A switch connects computers together inside the same network. It helps send data to the correct device.

What is an IP address?

An IP address is the computer's address on the network.

What is a subnet mask?

The subnet mask tells the computer which other IP addresses are in the same local network.

What is ping?

Ping checks whether another computer is reachable.

What is DHCP?

DHCP automatically gives IP addresses to computers, so we do not have to type them manually.

What is DNS?

DNS lets us use names instead of IP addresses. It changes a name like library_pc1 into an IP address like 192.168.2.100.

Why does ping fail for 192.168.1.15?

Because no device has that IP address. The computer sends a message, but nobody replies.

Why do all PCs need different IPs?

Because each computer needs its own unique address. If two computers share one address, the network gets confused.

9. Short checklist for you before the lab

Before students arrive

- Open Packet Tracer and confirm it works.
- Build Activity 1 once yourself.
- Build Activity 2 once yourself.
- Test manual IP ping in Activity 1.
- Test DHCP assignment in Activity 2.
- Test DNS name ping in Activity 2.
- Prepare a simple explanation of IP, switch, DHCP, DNS, and ping.
- Check whether students need to create images folders.
- Decide what DNS names students should use consistently.

During Activity 1

- 5 PCs and 1 switch are present.
- Straight-through cables are used.
- IPs are in the 192.168.1.x range.
- Subnet mask is 255.255.255.0.
- Ping works between existing PCs.
- Ping fails for the unused IP if tested.

During Activity 2

- 1 server, 1 switch, and 3 PCs are present.
- Server static IP is 192.168.2.2.
- DHCP service is ON and the pool is saved.
- PCs are set to DHCP and receive 192.168.2.100+ addresses.
- DNS service is ON.
- DNS names match actual PC IPs.
- Hostname ping works.

10. Missing or incomplete parts in the repo

DNS naming is inconsistent

The Activity 2 README uses different examples such as library_pc1, pc2_library, library.pc01, and library.pcXX. Best fix: use one consistent pattern, such as library_pc1, library_pc2, library_pc3.

Default gateway is included but no router exists

The DHCP settings include Default Gateway 192.168.2.1, but the topology does not include a router. This is not fatal because the lab focuses on local network communication.

Screenshots are requested, but image folders may not exist

The journal files refer to images/topology.png and similar paths. Students may need to create images folders manually.

DHCP DNS records depend on assigned IPs

If students turn on DHCP in a different order, PCs may receive different IPs. They should check each PC with ipconfig /all first.

Activity 2 step numbering skips from Step 3 to Step 5

This is minor and does not affect the lab.

Final practical advice for you

For Week 4, the biggest beginner confusion will probably be:

- IP addresses must be unique.
- Devices must be in the same network to ping each other directly.
- A switch connects devices but does not automatically give IP addresses.
- DHCP gives IPs automatically.
- DNS names must exactly match what was configured.
- If DNS ping fails, first test IP ping.

A good troubleshooting order to teach students:

```
1. Are the cables connected?
2. Are the link lights green?
3. Does the PC have an IP address?
4. Is the subnet mask correct?
5. Can it ping the server/same-network PC by IP?
6. If IP ping works but name ping fails, check DNS.
```

This order will help them debug most problems in the lab.